2026년 「차세대 반도체 글로벌 인재 육성 프로그램」 연수생 모집 공고

한국나노기술원은 반도체 분야를 선도할 우수 인재 육성을 위해, 「차세대 반도체 글로벌 인재 육성 프로그램」에 참여할 연수생을 다음과 같이 모집·선발하고자 합니다. 본 프로그램에 관심 있는 분들의 적극적인 참여를 바랍니다.

2026. 2. 19.

한국나노기술원 원장직무대행 박 노 재

1 개요

○ **(목적)** 국제교류를 통한 반도체 분야 글로벌 핵심 인재 발굴·지원하고 국제 경쟁력을 갖춘 전문인력으로 육성

○ **(내용)** 한국나노기술원-National Cheng Kung University(국립 성공대학교, 대만) 공동으로 제공하는 반도체 연수 프로그램 지원

○ **(연수기간)** 2026년 7월 20일 ~ 8월 7일(3주간)

○ **(모집인원)** O명

○ **주요일정**

일정	구분	교육내용
1주차 ('26.7.20~7.24.)	KANC	-기본 교육(반도체 공정 및 소자 교육, 안전 교육 등)
2주차 ('26.7.27~8.2.)	NCKU	(과정) Advanced semiconductor manufacturing and packaging/testing -해외 연계 교육(w.TSRI) -산업/연구기관 방문(ASML, Academia Sinica South Campus)
3주차 ('26.8.3~8.7.)		

2 지원대상 및 내용

○ **(지원대상)** 대한민국 국적을 보유한 자로서, 국내 대학 및 대학원 재적자 또는 졸업자

- 재적자 : 대학교 3학년 이상 ~ 대학원 석사과정* 1학년(재학생 또는 휴학생)

  * 석·박사 통합 과정 포함

- 졸업자 : 학사 학위 취득 후 1년 이내의 미취업자

○ **(지원내용)** 연수 프로그램 비용, 출입국 관련 비용(항공료 등), 체재비 지원

※ 세부 지원 항목은 별도 안내 예정

중도 포기 시, 연수에 소요된 비용을 전액 환수할 수 있음

3 신청안내

○ **(신청기간)** 2026년 2월 19일 ~ 3월 8일까지

○ **(신청방법)** 이메일(edu@kanc.re.kr) 접수

○ **제출서류**

구분	제출서류	제출시기	비고
1	[서식1]신청서	신청 시 제출	PDF
2	[서식2]연수계획서	신청 시 제출	
3	[서식3]개인정보 수집·이용 및 제3자 제공 동의서	신청 시 제출	
4	재학증명서 또는 졸업증명서*	최종 합격 시 제출	
5	외국어, 자격/면허, 수상, 교육/연수 증빙 서류	최종 합격 시 제출	

* 공고일 기준 3개월 이내 발급된 서류에 한하여 인정

4 선정절차 및 평가

○ **선정절차**

모집공고 및 접수 (‘26.2.19~3.8)		사전검토 (‘26.3.9~3.10)		선정평가 (‘26.3.11~3.20)		결과발표 (‘26.3.24)
지원자 모집 (신청서 제출 및 접수)	➢	지원자격 등 검토	➢	서류 및 면접평가 (영어면접 포함)	➢	선정 결과 안내 (개별)
신청자 → 한국나노기술원		한국나노기술원		선정평가위원회		한국나노기술원 → 선정자

※ 상기 일정은 운영 상황에 따라 변경될 수 있음

○ **(평가사항)** 서류평가와 면접평가로 구성

- 공정한 평가를 통한 연수생 선정을 위해 관련 분야 내·외부 전문가로 구성된 선정평가위원회 운영(면접 일정은 대상자에게 추후 공지)
- 평가항목

항목	내용	배점	비고
적합성	– 연구 프로그램의 목적 및 내용 이해도 – 전공 적합성 – 참여 동기 및 목표 등	40	서류/면접
역량	–반도체 관련 지식 및 이해 수준 – 문제 해결 능력 – 의사소통 능력	30	서류/면접
활용성	–연수 성과의 향후 진로 활용 가능성 –연수 종료 후 후속 활동 계획	30	서류/면접

※ 평가 결과의 세부 사항은 평가의 공정성 확보 및 개인정보 보호를 위하여 공개하지 않음

5 기타 안내

○ 연수생은 연수 종료 후 결과보고서를 제출(성과 발표 포함)해야 하며, 성과 확산 및 정보 제공을 목적으로 공개 가능

○ 연수생은 연수 일정에 따른 실제 연수 기간 종료 후 즉시 귀국하여야 하며, 별도의 추가 체류 불가

6 문의처

○ 한국나노기술원 나노인력양성실 담당자

- (전화) 031-546-6540 (이메일) edu@kanc.re.kr

첨부1 일정표(안)

Time Interval (UTC+8)	27–Jul	28–Jul	29–Jul	30–Jul	31–Jul	1–Aug	2–Aug
	Mon	Tue	Wed	Thu	Fri	Sat	Sun
9:10–10:00	Arrival in Tainan	Opening Ceremony (9:30)	②Semiconductor Materials	④Neural Networks	⑥Vacuum System	Cultural Exchange	Cultural Exchange
10:10–11:00		Campus Tour (10:30)					
11:10–12:00							
12:00–14:00		Lunch break	Joint Luncheon Talk	Lunch break	Joint Luncheon Talk		
14:10–15:00		①Quantum Applications of the defects in Semiconductor materials	③Applications of Numerical Simulation in the IC Packaging Industry	⑤IC Packaging Computer–Assisted Analysis Operation Industry	⑦Cleanroom Technology		
15:10–16:00							
16:10–17:00							

Time Interval (UTC+8)	3–Aug	4–Aug	5–Aug	6–Aug	7–Aug	8–Aug	9–Aug
	Mon	Tue	Wed	Thu	Fri	Sat	Sun
9:10–10:00	⑧Semiconductor Fabrication	⑨Nanophotonic Devices	⑪Worldwide Innovative 3DHI Packaging Technology Trend	⑬Nanophotonic Devices	Industry Visit (ASML)	Departure from Tainan	
10:10–11:00							
11:10–12:00							
12:00–14:00	Lunch break	Lunch break	Joint Luncheon Talk	Lunch break	Lunch break		
14:10–15:00	Institute Visit (Academia Sinica South Campus)	⑩Semiconductor Fabrication	⑫Lithography & Mask Technology	⑭Latest Innovations and Challenges of Semiconductor Technology for 3nm Node and Beyond	Final Presentation		
15:10–16:00							
16:10–17:00					Closing Ceremony Banquet		

첨부2 주요 프로그램(안)

①Quantum applications of the defects in semiconductor materials	Dr. Yu-Chen Chen(Academia Sinica)
This session will introduce the basic knowledge of various quantum technologies and focus on the quantum aaplications of the defects in semiconductors.	
②Semiconductor Materials	Prof. Chao-Hung Wang(NCKU)
This session introduces the fundamental properties and applications of semiconductor materials. Topics include crystal structures, electronic band theory, charge carriers, doping, and transport phenomena. The course also covers key semiconductor materials such as silicon, compound semiconductors, and emerging materials, along with their roles in electronic and optoelectronic devices. Emphasis is placed on linking material properties to device performance and modern semiconductor technologies	
③Applications of Numerical Simulation in the IC Packaging Industry	Dr. Li-Hsuan (Leo) Shen(CoreTech System)
This session provides an in-depth exploration of the challenges faced in IC packaging, focusing on innovative solutions by applying mechanical theory, material science, and numerical simulation. Participants will gain a conceptual understanding of how these disciplines interact to address and overcome industry-specific difficulties.	
④Neural Networks	Prof. Chao-Hung Wang(NCKU)
This session provides an introduction to neural networks and their fundamental principles. Topics include basic neuron models, network architectures, learning algorithms, and training methods. The course covers multi-layer perceptrons, convolutional neural networks, and deep neural networks, with applications in pattern recognition, signal processing, and artificial intelligence, emphasizing the connection between mathematical foundations and practical implementations.	
⑤IC Packaging Computer-Assisted Analysis Operation Course	Mr. Goran Liu(CoreTech System)
This session starts with the IC package 2D design drawing to construct a 3D mesh, sets the package material parameters and process conditions, and then performs filling, curing, and warpage analysis. The analysis results are used to illustrate common defects in packaging and to propose design change and improvement directions.	
⑥Vacuum System	Mr. Jacy Liao(Applied Materials)
This session is designed for those individuals in the semiconductor industry who require the knowledge and skills to work on the vacuum system. Agenda: (1) Basic Vacuum Theory (2) Vacuum System Overview (3) Vacuum Pump Introduction (4) Vacuum Gauge Introduction (5) Leak Check Technology	

⑦Cleanroom Technology	Mr. Jacy Liao(Applied Materials)
This session is designed for those individuals in the semiconductor industry who require the knowledge and skills to familiar with Cleanroom Technology. Agenda: (1) Semiconductor device scaling (2) Particle (3) Cleanroom overview (4) How to protect the devices on wafer–Wafer transfer/storage/shipping (5) Cleanroom Protocol–Gowning Practice.	
⑧Semiconductor Fabrication	Prof. Ying–Yuan Huang(NCKU)
Semiconductor Fabrication delves into the heart of the microelectronics industry, focusing on the manufacturing process of semiconductor devices. This session offers students an essential understanding of the steps involved in the fabrication of semiconductors, including material preparation, chemical/physical vapor deposition, photolithography, etching, and ion implantation. Students will gain insight into the sophisticated processes and innovations that enable the production of ever–smaller and more powerful semiconductor devices.	
⑨Nanophotonic devices	Prof. Jui–Nung Liu(NCKU)
This session provides an introduction to nanophotonics based on semiconductors and their device applications. Several important types of optical resonators will be introduced.	
⑩Worldwide Innovative 3DHI Packaging Technology Trend	Mr. Albert Lan(Applied Materials)
This session introduces the worldwide trends in innovative 3D heterogeneous integration (3DHI) packaging technologies. It covers (1) the development of C2W/W2W hybrid bonding for AI and HPC applications, and (2) the development of FO–PLP and advanced substrate technologies for next–generation AI/HPC packaging.	
⑪Lithography & Mask Technology	Dr. Frederick(Applied Materials)
As technology nodes continue to shrink, advanced lithography techniques play an increasingly essential role in shaping the speed, power efficiency, and complexity of today's chips. Photolithography is a key microfabrication technique that uses light, precision optics, mechanical control, and materials engineering to transfer circuit design patterns onto a wafer.	
⑫Latest Innovations and Challenges of Semiconductor Technology for 3nm Node and Beyond	Mr. Kevin Hsiao(Applied Materials)
In this session, the latest innovation of semiconductor process to drive the success of advanced technology nodes will be described, from transistor structure evolution, to impact and challenges of each process technologies including lithography/deposition / etch / CMP. Furthermore, the future challenges and opportunities beyond 3nm nodes to keep Moore’s Law alive, such as Gate–All–Around (GAA) structure, backside power rails, and chiplet heterogeneous integration will also be presented.	

첨부3 현장 견학(안)

□ Institute Visit(Academia Sinica South Campus)

Students will visit Academia Sinica, Taiwan’s premier national research institution dedicated to fundamental and interdisciplinary research. The visit focuses on facilities of the Research Center for Critical Issues (RCCI), which provides shared research platforms supporting advanced scientific and technological studies.

The program includes an introduction to the Quantum Optoelectronics Advanced Fabrication and Precision Analysis Facility, featuring advanced tools for device fabrication, materials processing, and high-precision characterization relevant to emerging electronic and photonic technologies. Students will also visit the Advanced Solar Cell Research and Measurement Facility, which supports solar cell development and materials analysis through comprehensive measurement and testing capabilities for energy-related research.

This visit offers a focused overview of how Academia Sinica’s research infrastructure supports frontier research and talent development in Taiwan.





Academia Sinica South Campus

☐ Industry Visit(ASML)

ASML is a global leader in advanced semiconductor lithography systems and a key enabler of cutting-edge chip manufacturing worldwide. The company is best known as the sole supplier of EUV (Extreme Ultraviolet) lithography systems, which are essential for advanced technology nodes and play a critical role in enabling innovation across AI, high-performance computing, and automotive electronics.

ASML places strong emphasis on engineering excellence, interdisciplinary collaboration, and long-term talent development, supporting a highly skilled workforce capable of managing complex, high-precision systems.

As part of the program, a brief company visit will be arranged to provide participants with a contextual understanding of ASML's operations and work environment, complemented by technical and career-sharing sessions led by professionals from diverse roles, with a focus on practical engineering work and real-world industry experience.







ASML